DEPARTMENT OF THE NAVY
CHIEF INFORMATION OFFICER
1000 NAVY PENTAGON
WASHINGTON, DC 20350-1000

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MEMORANDUM FOR DISTRIBUTION

Subj: DEPARTMENT OF THE NAVY GUIDANCE ON THE USE OF GENERATIVE ARTIFICIAL INTELLIGENCE AND LARGE LANGUAGE MODELS

Ref: (a) FACT SHEET: Biden-Harris Administration Secures Voluntary Commitments from Leading Artificial Intelligence Companies to Manage the Risks Posed by AI | The White House
(b) DoD Department of Defense Policy on the use of Classified Uncontrolled Information - DoDI 5200.48
(c) Security Policy for Generative Artificial Intelligence (AI) and Large Language Models (LLM), General Services Administration (GSA)
(d) DAF Use of Large Language Models, U.S. Air Force
(e) Security Implications of ChatGPT, Cyber Security Alliance (CSA)
(f) Generative AI: The Basics, Gartner

1. Purpose. The purpose of this memorandum is to offer interim guardrail guidance when considering the use of Generative Artificial Intelligence (AI) or Large Language Models (LLM).

2. Background. Generative AI models present unique and exciting opportunities for the Department of the Navy (DON). These models have the potential to transform mission processes by automating and executing certain tasks with unprecedented speed and efficiency. Other service components are already performing extensive experiments with this advancing technology. This has led to an extensive demand for their adoption across the DON. These powerful artificial intelligence-based algorithms have the extraordinary ability to return human-like responses to user-created prompts. They can produce new or novel responses derived from the corpus of data upon which they were trained. The output is not limited to just text, but can also be delivered in the form of images, audio, video, or other forms of unstructured data like electromagnetic signals. Some LLM variants are even being used by software engineers and developers to help generate first-draft (and iterative) code more quickly. This novel use is revolutionizing software engineering practices within the industry. Commercially available examples of these tools include, but are not limited to: OpenAI's ChatGPT, Google's Bard, and Meta's LLaMA. While offering great promise, LLMs warrant a cautious approach as indicated by the concern expressed by the National Command Authority, emerging technology industry leaders and academia.

3. Guidance.

a. General Use: Generative AI tools are not infallible and can produce "hallucinations" and biased results. Hallucinations are a phenomenon whereby responses include or incorporate fabricated data that appears authentic. For this, and many other reasons, these tools must be accompanied by a robust review process which includes the critical thinking skills of human expertise.

b. Military Use: Commercial AI language models are not recommended for operational use-cases until security control requirements have been fully investigated, identified and approved for use within controlled environments.

c. Data Protection and Security: It is imperative to understand that LLMs save every prompt they are given, which presents a persistent Operational Security (OPSEC) risk. Pre-existing established policy and guidance governs the use of sensitive information and appropriate use of Distribution A content. This policy extends to the use of unregulated AI LLM and includes source code as outlined in ref (b) (DoDI 5200.48). The use of proprietary or sensitive information poses a unique security risk and has the potential to lead to data compromise when employed by commercial generative AI models. The aggregation of individual generative AI prompts and outputs could lead to an inadvertent release of sensitive or classified information. In response to these security concerns, the DON is in the process of establishing rules of engagement and access to LLM technology through Jupiter, the DON Enterprise data and analytics platform. This will be accompanied by established security measures to ensure continued alignment with existing policies and the protection of government data.

d. Verification and Validation: In order to effectively leverage the complete potential of these tools, they must be complemented by human expertise. The development of specialized AI for government use necessitates a meticulous information validation process before model generation, while human proficiency remains indispensable. Present AI technology provides an imperfect rate of response without a method to ensure consistent and predictable output. The integration of AI should adhere to the core principles of Responsible Artificial Intelligence (RAI). This entails not only proofreading and fact-checking inputs and outputs for accuracy but also verifying the credibility of sources, rectifying factual inaccuracies, and actively addressing any potential violations of intellectual property rights. This verification process should be an integral aspect. Concurrently, it is of equal importance that all AI-generated software code undergoes a comprehensive review, evaluation, and testing procedure within a controlled, non-production environment before implementation. The incorporation of generative AI into any application warrants consideration of an Authority to Operate (ATO) assessment.

e. Responsibility and Accountability: The responsibility and accountability for user-induced vulnerabilities, violations and unintended consequences incurred by the use and adoption of LLMs ultimately resides with each individual organization's respective leadership.

4. Conclusion. The DON is committed to exploring how Generative AI can improve response delivery speeds and increase value to the Department of the Navy. Generative AI can be a force multiplier, but, as illustrated, it comes with inherent security vulnerabilities, risks and potential secondary consequences, which should be carefully considered while we learn how to employ them safely for the benefit of the DON's mission.

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Jane O. Rathbun
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